

AI-Based Quality Risk Prediction Report

Date: 20-07-2026

Time: 02:15 PM

Prediction Summary

Prediction	FAIL
Failure Risk	76.02%
Risk Level	High Risk
Likely Defect	OD Out of Tolerance
Confidence	93.70%

Input Parameters

Parameter	Value
Material	PU
Braid	None
Machine	EXT-01
Target OD (mm)	4.0
Target ID (mm)	1.5
Wall Thickness (mm)	1.25
OD Tolerance (mm)	0.1
ID Tolerance (mm)	0.1
Wall Tolerance (mm)	0.1
Vacuum (bar)	-0.4561
Screw Speed (RPM)	47.31
Zone 2 Temp (°C)	174.07
Zone 3 Temp (°C)	181.07
Melt Pressure (bar)	151.42
Puller Speed (m/min)	17.635

Material Moisture (%)	0.0917
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AI Recommendations

- Decrease Material Moisture from 0.09 to 0.08 %
- Increase Melt Pressure from 151.42 to 158.99 bar
- Increase Zone 2 Temperature from 174.07 to 182.77 °C
- Decrease Screw Speed from 47.31 to 44.94 RPM
- Increase Zone 3 Temperature from 181.07 to 190.12 °C

Generated by: AI-Based Thermoplastic Extrusion Quality Prediction System